

Uncertainty and failure analysis

Project: Image-pattern Network Investigation

1. Prepare the purpose, prerequisites and materials

Teach the learning cycle

Primary target: Distinguish a predicted class, confidence score and observed correctness

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Transfer learning

Without opening the old lesson, explain one key idea from Transfer learning. Give an example and a mistake that the idea helps you avoid.

Look for: Trace a reused feature extractor and a newly fitted output head

Reference example, not the only acceptable wording: Teacher explanation: The extractor gives $h = 6$ and the head predicts 1. Weights $[1, 1]$ and bias 0 stay frozen; only the threshold is fitted. h is a nonnegative sum of counts, with no probability mapping or calibration.

Embeddings and neighbourhoods

Without opening the old lesson, explain one key idea from Embeddings and neighbourhoods. Give an example and a mistake that the idea helps you avoid.

Look for: Look up a vector using a token ID and preserve sequence shape

Reference example, not the only acceptable wording: Teacher explanation: Vectors are $[[1, 1], [0, 1], [0, 2], [0, 0]]$, mask $[1, 1, 1, 0]$ and shape 4 by 2.

Training a small image model

Without opening the old lesson, explain one key idea from Training a small image model. Give an example and a mistake that the idea helps you avoid.

Look for: Trace a complete image model with fixed features and one trained weight

Reference example, not the only acceptable wording: Teacher explanation: $d = 2$, $h_1 = 2$ and $h_2 = 0$. Score is 0.9, prediction class 1 and squared error $(0.9 - 1)^2 = 0.01$.

Use two 45-minute sessions. These are supplied classifier outputs for arithmetic and reasoning, not calibrated probabilities from a trained model. Keep overall accuracy, accepted-case accuracy and coverage visible together. Do not imply that review guarantees correction or that model confidence diagnoses the cause of uncertainty.

Outcomes

Distinguish a predicted class, confidence score and observed correctness

Calculate accuracy and coverage for a review policy

Investigate failure patterns without claiming an untested cause

Preparation

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Prerequisite answer

Two constructed cases provide very limited evidence. New inputs can differ, and validation used to choose settings is not untouched final evaluation. Performance needs representative independent testing.

End goal for the pictorial case

By the end, I can explain uncertainty and failure analysis using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain and model the mechanism

A model can be confidently wrong. Use these supplied outputs for a fictional image classifier: class 1 means a triangle is present; class 0 means it is absent. The values p_1 are probability estimates for class 1, invented for this exercise rather than produced by a trained model. Predict 1 when $p_1 \geq 0.5$ and 0 otherwise. The selected-class confidence is $\max(p_1, 1 - p_1)$.

A predicts 1 with confidence 0.95 and is correct. F predicts 0 with confidence 0.90 and is wrong: $p_1 = 0.10$ is not low confidence in the selected class. Reference labels are supplied for this exercise; real investigations must check their quality too.

A score alone cannot establish reliability. Calibration asks whether, across many comparable cases, predictions assigned a probability such as 0.90 are correct about 90 percent of the time. One correct or incorrect case cannot answer that question.

The predictions for A through F are 1, 1, 1, 1, 0, 0. A, C and E are correct, so overall accuracy is $3 / 6 = 50$ percent. There are two false positives (B and D), one false negative (F), two true positives and one true negative. Different errors can matter differently; a single accuracy number hides that distinction.

The function returns overall accuracy, coverage and accuracy among accepted cases. It assumes numeric scores and binary labels. None means no cases were accepted, so accepted-case accuracy is undefined, not zero or perfect. Keep counts with proportions and declare the comparison at the boundary.

Term · Meaning

Term: Confidence score

Meaning: The model's score for its selected class, which may not match observed reliability

Term: Calibration

Meaning: Agreement between predicted probabilities and observed frequencies across many comparable cases

Term: Abstention

Meaning: Withholding an automatic prediction and following a declared alternative process

Term: Coverage

Meaning: The fraction of all cases on which a system makes an automatic prediction

Confidence does not remove every error

Cases · At cutoff 0.80 · Observed result

Cases: A, B, C, F

At cutoff 0.80: Accept automatically

Observed result: 2 correct, 2 wrong

Cases: D, E

At cutoff 0.80: Defer for review

Observed result: 1 correct, 1 wrong

The policy accepts confidence at least 0.80. Coverage is 4 / 6 and accepted-case accuracy is 2 / 4. The confident errors B and F remain.

Guided: only A is accepted. Coverage is 1 of 6; accepted accuracy is 1 of 1. This single constructed success, selected by a rule examined on these data, does not establish future reliability. Exit: report one correct automatic prediction out of one accepted, with five of six deferred and no reviewer outcomes. Require representative untouched evaluation, subgroup and calibration evidence, measured reviewer outcomes and workload, and a comparison with a declared baseline.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Confidence does not remove every error

Cases · At cutoff 0.80 · Observed result

Cases: A, B, C, F

At cutoff 0.80: Accept automatically

Observed result: 2 correct, 2 wrong

Cases: D, E

At cutoff 0.80: Defer for review

Observed result: 1 correct, 1 wrong

The policy accepts confidence at least 0.80. Coverage is 4 / 6 and accepted-case accuracy is 2 / 4. The confident errors B and F remain.

Deferred cases are not automatically repaired. A review workflow needs an assigned reviewer, suitable evidence, a decision record and an outcome check. Report review workload and the final system's results separately; we have no reviewer outcomes here. Do not erase deferred cases from the report or count them as correct.

Raise the cutoff to 0.95 for the same six cases. Which cases are accepted, what are coverage and accepted-case accuracy, and why does that result not guarantee perfect future predictions?

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger

claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Uncertainty and failure analysis

Keep positive-class probability separate from confidence in the chosen label.

The facts we will use

Supplied p(class 1): A: 0.9 reference 0; B: 0.6 reference 1; C: 0.2 reference 0. Predict class 1 at $p \geq 0.5$; confidence = $\max(p, 1 - p)$.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

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Positive probability is not always chosen-label confidence

	p(class 1)	prediction	confidence	reference
A	0.9	1	0.9	0
B	0.6	1	0.6	1
C	0.2	0	0.8	0

Read a cell using both its row and column labels.

Uncertainty and failure analysis

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Predictions

A predicts 1 wrong; B predicts 1 correct; C predicts 0 correct.



2. Selected-label confidence

Confidences are 0.9, 0.6, 0.8. C's confidence is 0.8 despite $p(\text{class } 1)=0.2$.



3. Selective review

Accept confidence ≥ 0.8 : A and C; accuracy 1/2 and coverage 2/3.

CHANGE ONE THING

Raise the acceptance cutoff to 0.85.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Predictions

What happens and why: A predicts 1 wrong; B predicts 1 correct; C predicts 0 correct.

Step: 2. Selected-label confidence

What happens and why: Confidences are 0.9, 0.6, 0.8. C's confidence is 0.8 despite $p(\text{class } 1)=0.2$.

Step: 3. Selective review

What happens and why: Accept confidence ≥ 0.8 : A and C; accuracy 1/2 and coverage 2/3.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Positive-class probability and confidence in the chosen label differ when the predicted class is zero. For C, a class-one probability of 0.2 gives class-zero confidence 0.8 under this toy rule. Selective accuracy counts only accepted cases; coverage counts how many were accepted. A stricter cutoff can retain a confidently wrong case and lower accuracy on this supplied set.

What this example does not establish

Supplied scores are not proven calibrated. Review policies need representative held-out evidence and clear handling of rejected cases.

Change one thing in the pictured case

Change: Raise the acceptance cutoff to 0.85.

Prediction: Only A is accepted: accuracy 0/1, coverage 1/3.

Reason: The most confident supplied case is wrong; a stricter cutoff is not guaranteed to improve this set's accuracy.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Time or stage · Teacher action

Time or stage: Session 1 start to 7 minutes

Teacher action: Retrieve the limits of the two-case validation result from Week 28. Define the triangle-present label.

Time or stage: 7 to 18 minutes

Teacher action: Trace A and F, then complete W1 with selected-class confidence distinct from p1.

Time or stage: 18 to 32 minutes

Teacher action: Build the error tally and complete W2. Trace the report function and its three returned quantities.

Time or stage: 32 to 45 minutes

Teacher action: Use the visual and W3 to retain deferred cases in workload reporting and handle an empty accepted set.

Time or stage: Session 2 start to 8 minutes

Teacher action: Complete the guided cutoff 0.95 case and critique a perfect-accuracy claim.

Time or stage: 8 to 20 minutes

Teacher action: Inspect the blur pattern and design a controlled comparison with source-grouped splits.

Time or stage: 20 to 35 minutes

Teacher action: Collect W4 independently, including denominators and the worse accepted-subset result.

Time or stage: 35 to 45 minutes

Teacher action: Use W5 and the exit response to evaluate causal and reliability claims. Give feedback, then the fresh retry.

Use the worked example in Sections 3 and 6. Collect a prediction before revealing its result; ask the learner to point to each intermediate fact.

Provide separate boxes for p_1 , predicted class, confidence, reference label and correctness. Use fractions before percentages. Extension: for A, B, C and F, mean confidence is 0.8875 while observed accuracy is 0.5. Discuss why this small grouped discrepancy motivates calibration study but cannot establish a population-wide calibration rate.

Reduce help deliberately

Completion task: For all six cases, list false positives and false negatives, count true positives and true negatives, and calculate overall accuracy. Why might a single accuracy hide useful information?

Teacher check: Teacher explanation: False positives are B and D; the false negative is F. A and C are the two true positives; E is the one true negative. Accuracy is 3 of 6, or 50 percent. It hides which errors occur

and how their consequences differ.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Uncertainty and failure analysis. Proposed explanation: “A low p_1 always means low confidence”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: For a binary model, $p_1 = 0.10$ selects class 0 with confidence 0.90.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

What is C’s positive-class probability and chosen-label confidence? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

$p(\text{class } 1)=0.2$; predicted class 0 has selected-label confidence 0.8. These quantities answer different questions. Changed-case check: Only A is accepted: accuracy 0/1, coverage 1/3. The most confident supplied case is wrong; a stricter cutoff is not guaranteed to improve this set’s accuracy.

Changed-case reasoning: Only A is accepted: accuracy 0/1, coverage 1/3. The most confident supplied case is wrong; a stricter cutoff is not guaranteed to improve this set’s accuracy.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the investigation and diagnose misconceptions

A low p_1 always means low confidence: For a binary model, $p_1 = 0.10$ selects class 0 with confidence 0.90.

Reviewing uncertain cases fixes all errors: B and F remain confidently wrong; deferred outcomes also need measurement.

A shared condition proves the cause: A small error pattern motivates investigation; label checks, controlled examples and independent evaluation are still needed.

Use Section 9 for the evidence ledger. Add a boundary or missing-input case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Guided and exit questions

Guided: only A is accepted. Coverage is 1 of 6; accepted accuracy is 1 of 1. This single constructed success, selected by a rule examined on these data, does not establish future reliability. Exit: report one correct automatic prediction out of one accepted, with five of six deferred and no reviewer outcomes. Require representative untouched evaluation, subgroup and calibration evidence, measured reviewer outcomes and workload, and a comparison with a declared baseline.

W1 Separate confidence from correctness

For B, E and F, give the predicted class, selected-class confidence and correctness. Explain why $p_1 = 0.10$ does not mean F has low confidence in its predicted class.

Teacher explanation: B predicts 1 with confidence 0.90 and is wrong. E predicts 0 with confidence 0.55 and is correct. F predicts 0 with confidence 0.90 and is wrong. F's low score for class 1 corresponds to a high score for class 0, the class selected.

W2 Count both types of error

For all six cases, list false positives and false negatives, count true positives and true negatives, and calculate overall accuracy. Why might a single accuracy hide useful information?

Teacher explanation: False positives are B and D; the false negative is F. A and C are the two true positives; E is the one true negative. Accuracy is 3 of 6, or 50 percent. It hides which errors occur and how their consequences differ.

W3 Audit the review policy

At cutoff 0.80, list accepted and deferred cases, coverage and accepted-case accuracy. Can you calculate final accuracy after human review from the supplied data? What does the report function return for accepted-case accuracy at cutoff 0.99?

Teacher explanation: Accepted cases are A, B, C, F; deferred cases are D and E. Coverage is 4 of 6 and accepted accuracy is 2 of 4. No reviewer outcomes are supplied, so final system accuracy is unknown. At 0.99 no case is accepted; accepted accuracy is undefined and the function returns None.

W4 Analyse new outputs independently

Four new cases have (p_1 , reference label): K (0.85, 0), L (0.70, 1), M (0.30, 0), N (0.05, 0). Give predictions and confidences, overall accuracy, and accepted cases at cutoff 0.80. Calculate coverage and accepted-case accuracy. Does this policy improve accuracy on the accepted subset here?

Teacher explanation: Predictions are 1, 1, 0, 0; confidences are 0.85, 0.70, 0.70, 0.95. L, M and N are correct, so overall accuracy is 3 of 4. K and N are accepted; coverage is 2 of 4 and accepted accuracy is 1 of 2. Accepted accuracy is lower than overall accuracy in this example.

W5 Design a failure investigation

All three blurred cases are wrong. A team claims blur caused the errors and plans to put clear images in training and their blurred copies in final evaluation. Identify the unsupported claim and the split problem; propose a better investigation.

Teacher explanation: Three failures show a pattern, not a proven cause. Related clear and blurred copies across splits can leak source-image information. Check labels and preprocessing, then hold related versions together in reserved source groups and compare controlled blur variants within those groups. Choose changes before new final evaluation and report the small sample limitation.

Independent assessment and fresh retry

W4 requires predictions 1, 1, 0, 0; confidences 0.85, 0.70, 0.70, 0.95; accuracy 3 of 4; accepted K and N, coverage 2 of 4 and accepted accuracy 1 of 2. Fresh retry: (p1, label) values (0.90, 1), (0.65, 0), (0.35, 1), (0.10, 0). Predictions are 1, 1, 0, 0; confidences 0.90, 0.65, 0.65, 0.90. Overall accuracy is 2 of 4. At cutoff 0.80 accept first and fourth: coverage 2 of 4 and accepted accuracy 2 of 2. Require a limitation despite the perfect accepted subset.

Assess independent transfer

Selected case: Four new cases have (p1, reference label): K (0.85, 0), L (0.70, 1), M (0.30, 0), N (0.05, 0). Give predictions and confidences, overall accuracy, and accepted cases at cutoff 0.80. Calculate coverage and accepted-case accuracy. Does this policy improve accuracy on the accepted subset here?

Selected case answer

Teacher explanation: Predictions are 1, 1, 0, 0; confidences are 0.85, 0.70, 0.70, 0.95. L, M and N are correct, so overall accuracy is 3 of 4. K and N are accepted; coverage is 2 of 4 and accepted accuracy is 1 of 2. Accepted accuracy is lower than overall accuracy in this example.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not

previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect the project, support and next lesson

Add a labelled uncertainty and failure analysis evidence sheet to Image-pattern Network Investigation. Show how you can distinguish a predicted class, confidence score and observed correctness. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week’s record; a matching answer without a trace is insufficient.

Provide separate boxes for p1, predicted class, confidence, reference label and correctness. Use fractions before percentages. Extension: for A, B, C and F, mean confidence is 0.8875 while observed accuracy is 0.5. Discuss why this small grouped discrepancy motivates calibration study but cannot establish a population-wide calibration rate.

Week 30 studies dataset shift, comparing development and new-use conditions without assuming a familiar confidence score implies familiar data.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-5/30, ai-5/32, ai-5/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.